References

1. Aluffi, P.A., Brandi, J., Bazzi, M., Kennedy, K., Arderne, M., Rodrigues, D. and Lotz, M. (2025) 'Categorising SME bank transactions with machine learning and synthetic data generation', *arXiv*. doi: 10.48550/arXiv.2508.05425.
2. Baker, S.R. and Kueng, L. (2022) 'Household financial transaction data', *Annual Review of Economics*, 14, pp. 47–67. doi: 10.1146/annurev-economics-051520-023646.
3. Bastian, G., Bianchi, I., Goldstein, M. and Montalvao, J. (2018) 'Short-term impacts of improved access to mobile savings, with and without business training: Experimental evidence from Tanzania', *CGD Working Paper*, 478.
4. Gayatri, Y.P.V. and Raju, A.N. (2026) 'Personal Finance Agent: Tracks Expenses, Categorizes Transactions, Suggests Budgets, and Alerts Overspending', *International Journal of Engineering Research and Science & Technology*, 22(2), pp. 1733–1742. doi: 10.62643/.
5. Zhang, Y., Garg, R., Golden, L., Brockett, P. and Sharma, A. (2023) 'Segmenting Bitcoin Transactions for Direction of Price Movement Prediction'. SSRN. doi: 10.2139/ssrn.4391916.
6. **Chatzigiannis, P., Gu, W.C., Raghuraman, S., Rindal, P. and Zamani, M. (2023), *Privacy-Enhancing Technologies for Financial Data Sharing*.** It is an arXiv preprint, arXiv:2306.10200.
7. **De Clercq, B. (2026). "The South African financial education ecosystem." *Journal of Financial Regulation and Compliance*, 34(3), pp. 416–437.** DOI: 10.1108/JFRC-05-2025-0130.
8. **Derbyshire, L.E. (2026). "Success factors for financial education targeted at small and micro-enterprise owners–a South African outlook." *Cogent Business & Management*, 13(1), Article 2603019.** The DOI: **10.1080/23311975.2025.2603019**.
9. **Derbyshire, L.E., Fouché, J.P. & McChlery, S. (2025). "Financial literacy education framework for small and micro-enterprise owners – a South African perspective." *Journal of Contemporary Management*, 22(2), pp. 235–272.** DOI: **10.35683/jcman1066.312**.
10. Dong, K., Jonnalagedda, P., Gao, X., Acharya, A., Kissa, M., Flores, M., Chawla, N.V. & Das, K. (2025). *Transaction Categorization with Relational Deep Learning in QuickBooks*. arXiv:2506.09234.
11. Duvenhage, C., Keyser, N. and Moepi, L. (2025) 'The impact of financial

literacy on savings in South Africa', *International Journal of Research in Business and Social Science*, 14(5), pp. 222–229. doi: 10.20525/ijrbs.v14i5.3985.

12. Ferilli, G.B., Palmieri, E., Miani, S. and Stefanelli, V. (2024) 'The impact of FinTech innovation on digital financial literacy in Europe: Insights from the banking industry', *Research in International Business and Finance*, 69, 102218. doi: 10.1016/j.ribaf.2024.102218.
13. García-Méndez, S., Fernández-Gavilanes, M., Juncal-Martínez, J., González-Castaño, F.J. and Seara, Ó.B. (2020) 'Identifying banking transaction descriptions via support vector machine short-text classification based on a specialized labelled corpus', *IEEE Access*, 8, pp. 61642–61655. doi: 10.1109/ACCESS.2020.2983584.
14. **Gilyard, S. (2026). "Budgeting for bills: The impact on daily spending." *Journal of Economic Behavior & Organization*, 244, 107495.** DOI: **10.1016/j.jebo.2026.107495**.
15. Granholt, P.K. and Nilsson, M. (2023) 'The neural net in spending', *DNB Tech*, 19 April.
16. **Jensen, R.I.T., Ferwerda, J., Jørgensen, K.S., Jensen, E.R., Borg, M., Krogh, M.P., Jensen, J.B. & Iosifidis, A. (2023). "A synthetic data set to benchmark anti-money laundering methods." *Scientific Data*, 10, 661.** DOI: **10.1038/s41597-023-02569-2**.
17. Jørgensen, R.K. and Igel, C. (2021) 'Machine learning for financial transaction classification across companies using character-level word embeddings of text fields', *Intelligent Systems in Accounting, Finance and Management*, 28(3), pp. 159–172. doi: 10.1002/isaf.1500.
18. Kotios, D., Makridis, G., Fatouros, G. and Kyriazis, D. (2022) 'Deep learning enhancing banking services: A hybrid transaction classification and cash flow prediction approach', *Journal of Big Data*, 9(1), 100. doi: 10.1186/s40537-022-00651-x.
19. Lesner, C., Ran, A., Wang, W. and Rukonic, M. (2020) 'Large-scale personalized categorization of financial transactions', *AI Magazine*, 41(3), pp. 63–77. doi: 10.1609/aimag.v41i3.5319.
20. Mbukanma, I. and Rena, R. (2022) 'Significance of financial literacy on personal savings behaviour: A study of the City of Tshwane and Mahikeng Municipality employees', *African Journal of Development Studies*, SI(2), pp. 189–206. doi: 10.31920/2634-3649/2022/SIv2a11.
21. Mbatane, S. and Kekana, M.K. (2025) 'Digital financial literacy among university students in South Africa: A quantitative study', *International Journal*

of Research in Business and Social Science, 14(7), pp. 249–267. doi: 10.20525/ijrbs.v14i7.4284.

22. Shuttleworth, C.C., Moolman, J., Basson, A. & Risenga, A. (2026). "Information and Communication Technology: A Gate or Gateway to Advance Financial Literacy." *International Journal of Economics and Financial Issues*, 16(2), pp. 1–8.

23. Munongo, S. and Pooe, D. (2025) 'The influence of small and medium-sized enterprise financial literacy on FinTech adoption in a fourth industrial revolution era', *South African Journal of Economic and Management Sciences*, 28(1), a6246. doi: 10.4102/sajems.v28i1.6246.

24. Mvunabandi, J.D. and Gadzikwa, L. (2024) 'A cognitive analysis of consumer literacy training and financial capability improvement among South Africans: A cross-sectional study', *Oradea Journal of Business and Economics*, 9(1), pp. 44–62. doi: 10.47535/1991ojbe181.

25. Nanziri, E.L. and Leibbrandt, M. (2018) 'Measuring and profiling financial literacy in South Africa', *South African Journal of Economic and Management Sciences*, 21(1), a1645. doi: 10.4102/sajems.v21i1.1645.

26. Olafsson, A. and Pagel, M. (2018) 'The ostrich in the household: Savings, financial literacy, and financial well-being'.

27. **Oliveira, G.P., Gertrudes, J.C. & Oliveira, R.B. (2023). "Spending pattern visualization using unsupervised machine learning."** *Proceedings of the 38th Brazilian Symposium on Databases*, pp. 167–176. DOI: **10.5753/sbbd.2023.231577**.

28. Ortiz, J.M., Teixeira, L.I., Falcão, N.N.L., Soki, E.A. and Almeida, R.M. (2024) 'Information simplification and default choices improve financial decisions: A credit card statement experiment', *Journal of Behavioral and Experimental Economics*, 110, 102193. doi: 10.1016/j.socec.2024.102193.

29. **Shuttleworth, C.C., Moolman, J., Basson, A. & Risenga, A. (2026). "Information and Communication Technology: A Gate or Gateway to Advance Financial Literacy."** ***International Journal of Economics and Financial Issues*****, 16(2), pp. 1–8.** DOI: 10.32479/ijefi.21806.

30. **Steinert, J.I., Cluver, L.D., Meinck, F., Doubt, J. & Vollmer, S. (2018). "Household economic strengthening through financial and psychosocial programming: Evidence from a field experiment in South Africa."** ***Journal of Development Economics*****, 134, pp. 443–466.** DOI: **10.1016/j.jdeveco.2018.06.016**.

31. Struwig, J., Roberts, B., Gordon, S., Clarke, L. and Laubscher, C. (2022) *Rands and sense: Financial literacy in South Africa: Briefing report 1*. Pretoria: Human Sciences Research Council.

32. Toran, L., Van Der Walt, C., Sammarone, A. and Keller, A. (2023) 'Scalable and weakly supervised bank transaction classification', *arXiv*. doi: 10.48550/arXiv.2305.18430.

33. **Tshaka, M., Simatele, M. & Copestake, J. (2026). "Beyond Financial Market Dualism: An Empirical Analysis of Variations in Use of Financial Services in South Africa."** ***Journal of Risk and Financial Management*****, 19(1), 47.** DOI: **10.3390/jrfm19010047**.

34. Roberts, B., Struwig, J., Gordon, S. and Zondi, T. (2021) *Financial Literacy in South Africa: Results from the 2020 Baseline Survey*. Pretoria: Financial Sector Conduct Authority (FSCA).

35. **Wang, L., Cheng, Y., Xiang, A., Zhang, J. and Yang, H. (2024) 'Application of Natural Language Processing in Financial Risk Detection',** ***arXiv*****. doi: 10.48550/arXiv.2406.09765.**

36. Haas, N., Horesh, Y., Shahar, S. and Resheff, Y.S. (2018) 'Identifying Financial Institutions by Transaction Signatures', *Proceedings of the First Workshop on Data Science in FinTech (FinTech@KDD'18)*, pp. 1–4. doi: 10.48550/arXiv.1808.00151.

37. **Nguyen-Cousins, T. and Upton, M. (2025) 'Determinants of financial well-being among online financial education learners: a mixed-methods study',** ***Discover Psychology*****, 5, 99. doi: 10.1007/s44202-025-00385-w.**

38. Yakubu, M., Anazodo, U., Adewole, M., Barfoot, T., Lee, T., Vercauteren, T., Shapey, J., King, A.P. and Hammers, A. (2025) 'Themed Challenges to Solve Data Scarcity in Africa: A Proposition for Increasing Local Data Collection and Integration', *arXiv*. doi: 10.48550/arXiv.2508.00925.

39. Warikandwa, T.V. (2021) 'Personal Data Security in South Africa's Financial Services Market: The Protection of Personal Information Act 4 of 2013 and the European Union General Data Protection Regulation Compared', *Potchefstroom Electronic Law Journal*, 24, pp. 1–32. doi: 10.17159/1727-3781/2021/v24i0a10727.

40. Lefait, G. and Kechadi, T. (2010) 'Customer Segmentation Architecture Based on Clustering Techniques', in *Proceedings of the Fourth International Conference on Digital Society (ICDS 2010)*, pp. 243–248. doi: 10.1109/ICDS.2010.47.